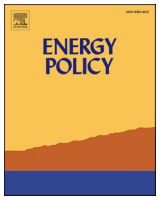
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Risk dynamics and strategies of China's solar PV industry chain under trade frictions: A review

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ABSTRACT

Amid intensifying global energy risks and trade frictions, China's leading solar photovoltaic (PV) industry chain faces significant challenges. This study follows the framework of risk management theory to systematically analyze the risk dynamics and strategies within China's solar PV industry chain under trade frictions. Firstly, the study identifies six categories of trade frictions risks, as well as internal risks across upstream, midstream, and downstream segments of the PV industry chain. Secondly, risk analysis is used to explore the impact of trade barriers on the industry and examines the interdependencies among internal risks, followed by a systemic coupling analysis of the entire PV chain. Thirdly, as traditional static risk assessment has limitations in evaluating the dynamic risks of China's solar PV industry chain, a dynamic risk assessment model is constructed that incorporates both temporal and spatial perspectives. Finally, China's strategies at different stages of industry development to respond to risks, including market diversification, domestic demand expansion, and technological innovation are reviewed. The review aims to provide theoretical guidance and practical insights for enhancing the risk management capacity of China's PV industry in an increasingly complex global environment.

1. Introduction

As global consensus grows on the need to reduce greenhouse gas emissions and increase the use of non-fossil energy, the development and application of clean energy technologies have become crucial competitive fields for nations, reshaping the global energy structure in profound ways (Zhang et al., 2022). Among these technologies, solar PV technology stands out as a cornerstone of sustainable energy, playing a pivotal role in the transition to a low-carbon future (Breyer et al., 2017). As the largest producer and consumer of solar PV products, China occupies a central role in the global PV landscape (Shuai et al., 2018; Zhu et al., 2019). China's solar PV industry has developed rapidly, with a gradually improving industry chain that covers raw material extraction, component manufacturing, and end-use applications (Zhao et al., 2015). The solar PV chain includes the mining and processing of materials such as silicon, intermediate manufacturing of cells and modules, system integration, and recycling at the end of the product lifecycle (Zhao et al., 2023a, 2023b). According to the National Energy Administration's report on PV power construction, the country added 216.3 GW of new grid-connected capacity in 2023 (National Energy Administration, 2024). Moreover, solar PV has surpassed hydropower to become the second-largest source of installed power capacity in China (National Energy Administration, 2025), further elevating its role in the country's energy structure.

However, global industry and supply chains are undergoing significant adjustments, with increasing uncertainties and challenges to economic globalization (Wang, 2020). Empirical evidence suggests that the Chinese PV industry is increasingly subject to external uncertainties driven by trade frictions (Meckling and Hughes, 2018; Xia et al., 2019; Wang et al., 2021a, 2021b). When trade imbalances, policy differences, conflicts of interest, or other reasons lead to disputes, barriers, or restrictions between two or more countries in international trade activities, trade frictions occur. (Liu et al., 2020). According to Gao and Song (2025), trade frictions against China's PV products increased in a

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fluctuating manner between 2009 and 2023, with the event count increasing from 19 in 2009 to 116 in 2023, and the number of countries and regions involved as initiators increasing from 19 to 278 over the same period. As the frequency of trade frictions has gradually escalated, academia has also investigated the impacts generated by trade barriers. Fang (2020) found that several countries, including the UK, Canada, India, and Turkey, imposed anti-dumping measures on Chinese PV exports. These measures not only led to a sharp decline in overseas orders but also triggered a chain reaction in the domestic market, including supply-demand imbalances and falling prices. Meanwhile, McCarthy (2016) analyzed the impact of European trade barriers on China's solar PV industry using a sample of 454 listed PV manufacturing enterprises, finding that trade barriers resulted in an average loss of 0.08 % for European firms compared to a loss of 1.6 % for Chinese firms. Similarly, research on China-US trade conflicts showed that these frictions have led to reduced energy demand and an overall economic slowdown in both countries, with China being hit harder (Xia et al., 2019; Liu et al., 2020). Although China and the United States (U.S.) exhibit strong trade complementarities in the solar PV supply chain, trade frictions pose a threat to the stability of the global supply chains (Wang et al., 2020). From an industry chain perspective, trade frictions have affected different segments of the PV industry chain in distinct ways. Due to the concentration of international raw material supply, price volatility in upstream materials has increased (Dustin and Morgan, 2023). In the midstream, export restrictions on manufacturing have led to overcapacity in solar cells and modules, pushing prices close to or even below production costs (Zhu, 2024). In the downstream, the reduction in overseas orders has continued to impact the domestic market, prompting Chinese solar PV enterprises to seek alternative markets (Zhu et al., 2024). The multi-level and multi-dimensional impacts make it difficult for traditional static risk assessment methods to fully capture the dynamic transmission of risks, thereby hindering the stable development of the industry.

Facing external pressures and an increasingly complex global risk landscape, China's solar PV industry has been exploring response strategies. In terms of policy orientation, the Chinese government has transitioned from early "export-oriented subsidies" to a model centered on "domestic demand stimulation" (Chen, 2015; Wen et al., 2020), which has shifted subsidies from the production side to the consumption side. For instance, feed-in tariff (FIT) mechanisms have been proven to stimulate domestic PV applications (Grau et al., 2012), increase investment in PV projects (Lin and Xie, 2024), and promote the growth of domestic solar installation capacity (Wang et al., 2016), thereby providing institutional support. As industry maturity has advanced, the policy framework has further evolved into a "subsidy phase-out" mechanism (Chen and Wang, 2022; Lin and Zhang, 2025), driving the industry's transition from policy dependence toward market competitiveness (Yan et al., 2019). From a technological perspective, Chinese solar PV enterprises have gradually achieved technological catch-up through patent collaboration networks, particularly in solar modules and first-generation solar cells (Dominguez Lacasa and Shubbak, 2018). Regarding industrial clustering, these firms have enhanced their R&D capabilities through local cooperative networks and regional industrial clusters. Technological coordination centers have also emerged in regions such as the Yangtze River Delta and the Pearl River Delta (Hu et al., 2024). In terms of market expansion, firms have taken steps to set up production facilities and project bases in Belt and Road Initiative (BRI) countries and in Europe. They aim to reduce market risks and help promote technology transfer and localization abroad (Jackson et al., 2021).

Thus, the purpose of this study is to analyze the dynamic risks faced by China's PV industry chain under trade barriers and explore response strategies. To achieve this goal, the study addresses the following three key questions: (1) What types of risks have emerged across different segments of China's solar PV industry chain during its development, particularly under the trade frictions? How have these risks changed over time? (2) Why is it necessary to assess dynamic risks, and how can they be evaluated? (3) What strategies has China adopted in response to these dynamic risks? To answer these questions, the study is structured into four main parts. First, it identifies the trade barrier risks and internal risks across the upstream, midstream, and downstream segments of China's PV industry chain at different stages of development. Second, it analyzes the impacts of trade barriers on the industry chain and examines the interactions among internal risk factors, followed by a coupling analysis of the overall system risks. Third, it reviews current risk assessment methods used in the PV sector, discussing their strengths and limitations, and proposes a dynamic risk assessment model based on the findings. Finally, it reviews the strategies developed by China's solar PV sector under external trade frictions at different stages.

The study makes three main contributions: (1) it systematically identifies various types of external trade barrier risks and categorizes internal risks within China's solar PV industry chain, including those related to resources, technology, policy, environment, market and finance. It further analyzes the interaction between external trade barriers and internal risks; (2) it moves beyond the limitations of traditional static risk assessment approaches by incorporating both temporal evolution and spatial transmission dimensions. Based on this, the study develops a dynamic risk transmission model to more comprehensively capture the evolving nature of risks and their transmission across different segments of the industry chain.

2. Material and methods

This study employs a comprehensive approach that integrates content analysis and bibliometric analysis to systematically examine the risk dynamics of China's solar PV industry chain amid trade frictions. The bibliometric analysis identifies research trends and key hotspots across different developmental stages from a quantitative perspective, while the content analysis traces the evolution of China's PV industry chain and its risks and challenges by a qualitative approach.

2.1. Criteria for inclusion and exclusion

To ensure the relevance and comprehensiveness of the literature selection, the following criteria for inclusion and exclusion were established.

(1) The keywords were designed to cover the entire solar PV industry chain, including upstream (e.g., silicon, polysilicon, wafers), midstream (e.g., solar cells, modules, panels), and downstream systems to ensure that the analysis reflects trade barriers' impacts across all stages rather than focusing solely on final solar PV products.
(2) Considering the various forms of international trade frictions, the search terms included both general descriptors ("trade friction", "trade dispute", etc.) and specific measures ("import tariffs", "export controls", "state subsidies", "technical barriers", "public procurement", "domestic content policy", "anti-dumping", "countervailing measures" and "anti-circumvention"). The goal was to capture a comprehensive set of literature on different types of trade barriers affecting the PV sector.
(3) Since the research focuses on dynamic risks in China's PV industry, "China" and "Chinese" were used as mandatory geographical filtersto exclude studies unrelated to the Chinese market.
(4) Studies were excluded if they were conference papers, preprints, and early access articles that had not undergone peer review, in order to ensure the authority and reliability of the included literature.

The terms used in the topic search were ("solar trade" OR "photovoltaic trade" OR "solar industry" OR "photovoltaic industry" OR "solar

sector" OR "photovoltaic sector" OR "solar market" OR "photovoltaic market" OR "solar export*" OR "solar import*" OR "photovoltaic export*" OR "photovoltaic import*" OR "PV industry" OR "PV market" OR "polysilicon" OR "solar wafer*" OR "photovoltaic wafer*" OR "solar cell*" OR "photovoltaic cell*" OR "PV cell*" OR "solar module*" OR "photovoltaic module*" OR "PV module*" OR "solar panel*" OR "photovoltaic panel*" OR "PV panel*" OR "solar system*" OR "photovoltaic system*" OR "PV system*") AND ("trade friction*" OR "trade dispute*" OR "trade conflict*" OR "trade barrier*" OR "trade war*" OR "trade risk*" OR "anti-dumping" OR "countervailing" OR "import tariff*" OR "import tax*" OR "import restriction*" OR "export restriction*" OR "export control*" OR "state aid" OR "subside*" OR "trade remedy*" OR "anti-circumvention" OR "domestic content policy" OR "public procurement" OR "technical barrier*" OR "technical trade barrier*" OR "TBT" OR "protectionism" OR "trade polic*") AND (China OR Chinese). The literature search was conducted primarily across the Web of Science (WOS) Core Collection, covering the period from January 1, 2002, to May 8, 2025. For papers with uncertain relevance, we conducted full-text reviews to make final inclusion decisions. Using the search strategy, the study reviewed a total of 207 research articles, review articles and book chapters.

2.2. Keyword co-occurrence analysis

Based on the keyword co-occurrence network generated using VOSviewer (see Fig. 1), the study conducts content clustering and temporal evolution analysis of the literature related to trade frictions in China's solar PV industry chain. The bottom-right corner of Fig. 1 displays a time axis. It represents the average year when each keyword appeared, with different colors indicating temporal progression. They help trace how the research focus has evolved over time.

Fig. 1 shows that each period has distinct keyword hotspots and stage characteristics. From 2012 to 2018, research mainly focused on fundamental elements such as industrial policy, energy policy, power generation capacity, and feed-in tariff mechanisms. The stage was marked by the early and rapid development of China's solar PV industry. Policy support played a key role in laying the sector's foundation (Zhang and He, 2013; Zhao et al., 2013; Zhang et al., 2014; Zhang, 2016a, 2016b). As China's PV products began to enter international markets on a large scale, initial solar trade frictions began to emerge (Hughes and Meckling, 2017). Between 2018 and 2020, the research focus shifted toward renewable energy innovation and technological advancement. Keywords such as renewable energy, cost reduction, technology development, and innovation became prominent. The shift reflects the industry's transition from being policy-driven to more market-oriented (Dominguez Lacasa and Shubbak, 2018; Urban et al., 2018; Xin-gang and You, 2018). During this period, technological breakthroughs and cost optimization helped China's PV products gain international competitiveness (Shuai et al., 2018; Wang et al., 2019; Zhu et al., 2019), which further intensified global trade frictions (Meckling and Hughes, 2018; Chen, 2019; Shum, 2019). Since 2020, trade-related issues have drawn significant attention. Keywords such as trade protection, protectionism, government subsidies, and supply chain coordination have become increasingly prominent. These studies respond to escalating international trade tensions affecting China's solar PV exports (Fang, 2020; Zhu et al., 2021a; Dehghanimadvar et al., 2024; Gao and Song, 2025; Lin and Liu, 2025).

Although the keyword analysis reveals the evolving trends and thematic focus of research on trade frictions related to China's solar PV industry chain, there are certain limitations. Most studies focus on broad topics such as energy policy and technological applications rather than specific risk factors, particularly those related to trade barrier contexts, which limits their visibility in the co-occurrence network. As a result, the keyword co-occurrence map tends to reflect the macro-level evolution of research themes, rather than offering a comprehensive picture of the dynamic risks of China's solar PV industry chain under trade frictions. Therefore, building on the bibliometric analysis, the study will further integrate qualitative content analysis to identify and categorize the risk factors of China's solar PV industry chain.

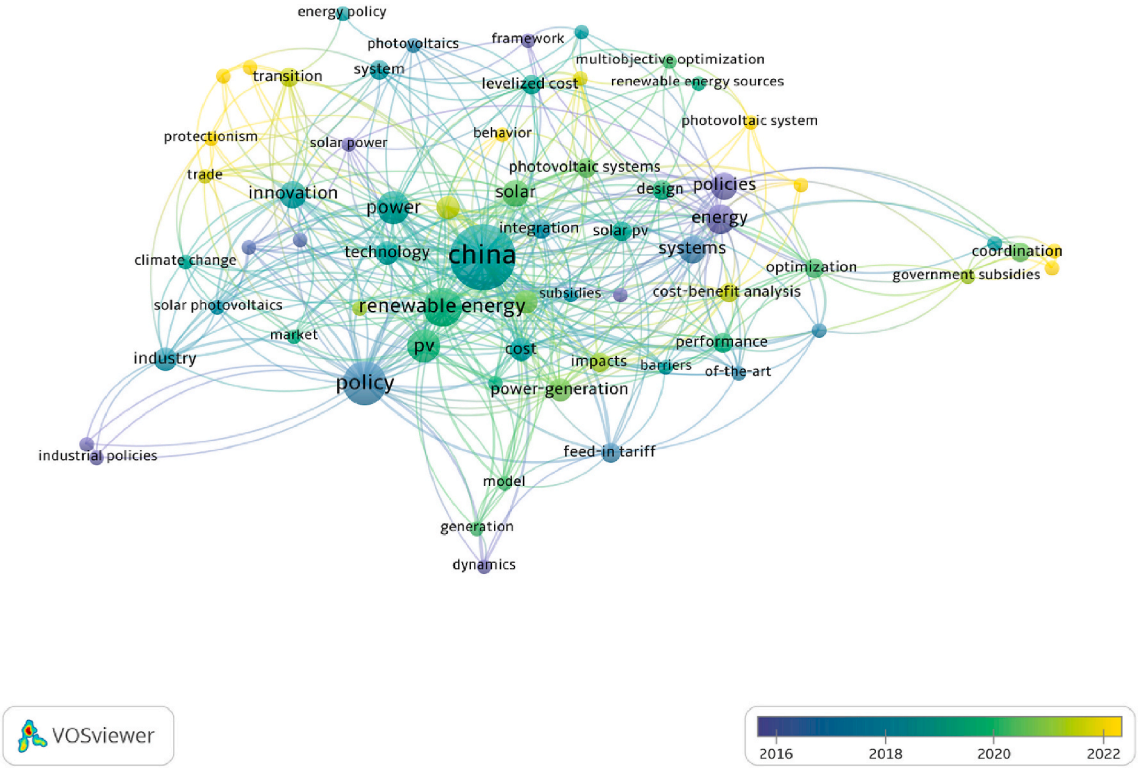


Fig. 1. Keyword co-occurrence network and temporal evolution.

3. Identifying risk factors of China's solar PV industry chain

3.1. Identification and classification of trade friction risks in China's solar PV industry chain

Based on data from the Global Trade Alert (GTA) database and existing literature (Kinzius et al., 2019; Gao and Song, 2025), trade frictions can be broadly classified into the following categories: (1) import tariffs; (2) other forms of import restrictions; (3) government aid and subsidy measures; (4) trade defense instruments; (5) export controls; (6) public procurement and localization policies; (7) technical barriers to trade (TBT). These measures have affected different parts of China's solar PV industry chain in different ways.

In the upstream and midstream segments, trade frictions mostly center on disputes over government assistance and subsidy measures. Other countries regard the concentration of China's solar PV manufacturing as a competitive threat, attributing it to what they perceive as unfair trade practices and state-led policies (Helveston et al., 2022). They argue that China's solar PV industry benefits from long-term government support through subsidies, credit access, and preferential land-use policies, which create a "price advantage" rather than a "technological advantage"(S. Zhao et al., 2023). Trading partners see these policies as distortions to fair market competition, triggering anti-dumping claims and trade disputes between exporting and importing countries in the solar sector (Chen and Su, 2018). In response, the US and some European countries have implemented a range of trade defense instruments to restrict market access for Chinese solar cells and modules. These measures include:

Trade defense instruments: Since 2012, the U.S. has imposed anti-circumvention practices (ACPs) on Chinese solar modules to offset the effects of government subsidies on export pricing (Abboushi, 2014). Between November 2011 and June 2024, the U.S. Department of Commerce launched two rounds of anti-dumping and countervailing duty investigations on Chinese crystalline silicon PV cells. In its second sunset review in 2024, the Department concluded that without countervailing duties, Chinese products would likely continue or resume being exported to the U.S. market at subsidy rates ranging from 25.56 % to 26.75 % (Ministry of Commerce PRC, 2024).

Import tariffs: Since 2018, under Section 301 of the Trade Act of 1974, the U.S. has imposed tariffs on Chinese imports in response to concerns about technology transfer and intellectual property violations. These tariffs cover a wide range of solar products, including modules, inverters, junction boxes, and backsheet materials, effectively targeting the entire PV supply chain. Combined with Section 201 tariffs imposed on all imports regardless of origin, these measures have sharply reduced China's direct exports of solar cells and modules to the U.S. (InfoLink, 2024).

Other forms of import restrictions: In 2013, the EU imposed its first round of anti-dumping and countervailing duties on Chinese PV modules, cells, and wafers. To resolve the dispute, the EU and China reached a Minimum Import Price (MIP) agreement, which included a quota system that limited annual sales volume. Companies participating in the MIP were exempted from punitive duties, while those outside the agreement faced anti-dumping duties of up to 47.6 % (Xinhua News Agency, 2013). The MIP arrangement was not a formal tariff, but effectively acted as a quantitative import restriction (McCarthy, 2016).

Trade defense measures have weakened the price competitiveness of Chinese solar cells and modules in the U.S. and European markets (Dehghanimadvar et al., 2024) and have indirectly accelerated the relocation of manufacturing and export activities to Southeast Asia (Lin and Liu, 2025). In August 2023, the U.S. Department of Commerce concluded a 16-month anti-circumvention investigation into solar products originating from Vietnam, Thailand, Malaysia, and Cambodia (International Trade Administration, 2023). As an extension of conventional trade remedies, anti-circumvention investigations aim to prevent firms from evading existing trade measures by routing products through third countries (Crochet and Zhou, 2023).

In the downstream segment, covering solar PV system integration, project implementation, and electricity sales, trade frictions are primarily reflected in public procurement and localization policies, as well as TBT. For example, India has required requires solar projects to use locally made components to safeguard local employment and industrial security, thereby excluding Chinese firms from participating in system integration activities (Johnson, 2016; Probst et al., 2020). Similarly, the U.S. enacted the Inflation Reduction Act (IRA) in 2022 and introduced domestic content requirements for federal tax incentives, which effectively limits the participation of Chinese manufacturers in downstream integration and project development (Leonelli and Clora, 2024). Such non-tariff barriers do not rely on tariffs. Instead, they block access through standards, certifications, and regulations. These rules create major challenges for Chinese companies trying to expand downstream (Chen, 2015). TBT also represents a form of implicit trade friction. Although solar PV products are not classified as agricultural goods, TBT measures are frequently used as technical restrictions that hinder the entry of Chinese products into high-end markets, such as product certification requirements and grid connection standards (Kelly and Mahesh, 2017).

In summary, China's solar PV industry faces very different types of trade frictions across different segments. The upstream and midstream segments encounter traditional trade defense measures such as anti-dumping, countervailing duties, and import licensing requirements. Moreover, the downstream segment is mainly challenged by non-tariff barriers, including localization policies and technical standards.

3.2. China's solar PV industry chain internal risks identification

To systematically identify the major internal risks within China's industry chain, the study summarizes and analyzes the relevant risks and their types across the upstream, midstream, and downstream segments, based on existing literature. To categorize the identified risks, we classify them into six dimensions based on their nature and impact areas. Resources dimension refers to risks related to the availability, supply, and pricing of raw materials and critical minerals essential for PV production; Technology dimension encompasses risks associated with technical capabilities, R&D investment, equipment dependency, and infrastructure constraints; Finance dimension covers risks related to production costs, investment requirements, profitability, and price pressures; Environment dimension includes risks stemming from pollution, waste management, and climate-related factors; Market dimension addresses risks concerning demand-supply imbalances, market concentration, competition intensity, and export dependency; Policy dimension involves risks related to government regulations, subsidies, and policy changes. This study organizes the relevant literature in chronological order of publication, with key details summarized in Table 1.

4. Risk analysis of China's solar PV industry chain

4.1. Risk impact analysis of trade frictions

As one of the most traditional instruments of trade protection, tariffs are widely employed in the global solar PV market, exerting profound impacts on industrial structures and corporate decision-making (Curran, 2015; Meckling and Hughes, 2018; Dong and Kouvelis, 2019). At the macro level, tariff barriers disrupt the global distribution of comparative advantages in solar PV products and drive the restructuring of global value chains. Castellanos et al. (2018) showed that tariffs substantially raise the minimum sustainable price of solar PV manufacturing, alter the previously optimal locations for production, and increase end-market prices, particularly affecting the global configuration of midstream and downstream segments of the industry chain. At the meso level, trade barriers have also reshaped the global market layout and export patterns of China's solar PV industry. Empirical evidence indicates that China's

Table 1
Risk identification of China's PV industry chain.

Risk segments	Upstream risk factors	Risk dimensions	Reference
Upstream	Dependence on foreign imports for key resources	Resources	Jiang (2013)
	High technical difficulty in silicon raw material production	Technology	Ye and Zhang (2014)
	High production costs for silicon raw materials	Finance	Ye and Zhang (2014)
	Environmental pollution caused by polysilicon enterprises	Environment	Liu et al. (2015)
	Over-concentration of polysilicon and wafer production	Market	Goldthau and Hughes (2020)
	Shortage of critical mineral supplies	Resources	(Behar et al., 2021; Zhang et al., 2021a 2021b; Gervais et al., 2023; Zhao et al., 2023a, 2023b)
	Raw material price fluctuations		
	Dependence on imported high-end manufacturing equipment	Technology	(Sekar et al., 2024; Xu et al., 2022)
Middle Stream	Low average value of patented inventions in China's PV cell and module sectors	Technology	de la Tour et al. (2011)
	Lower R&D investment ratio compared to Western firms		
	Dependence on financing for solar panel procurement	Finance	Sun et al. (2014)
	Domestic demand for PV cells and modules is lower than international demand	Market	Wang et al. (2014)
	High export dependency; sluggish global demand has led to domestic overcapacity	Market	Wang et al. (2014)
	Increasing pressure on PV cell and module prices	Finance	Wang et al. (2014)
	High market concentration in PV cell and module industries	Market	Ladislaw et al. (2021)
	Intensified market competition	Market	Zhang et al. (2023)
	Profitability risks for PV enterprises	Finance	Yang et al. (2024)
Down Stream	High initial and maintenance costs, along with long investment payback periods	Finance	Zhang et al. (2012)
	Severe imbalance between PV installation capacity and grid absorption capacity	Technology	Xu (2016)
	Reduction in government subsidies	Policy	Yan et al. (2019)
	Pollution from retired PV systems	Environment	Pu et al. (2021)
	Extreme weather events pose constraints on the effective utilization of solar energy	Environment	Sun et al. (2024)

PV exports to anti-dumping imposing countries declined sharply in 2011, while exports to other regions increased substantially (Jia et al., 2016; Kuang and Xiang, 2019). Zhu et al. (2021a) quantitatively verified that import restrictions imposed by the U.S. and the EU triggered a "diversion effect" in export destinations since 2011, and found that, with approximately 22.7 % of exports from 2012 to 2016 shifting from Western markets to emerging economies not subject to tariffs, such as those in Latin America, the Middle East, and Africa. Furthermore, Zhu et al. (2021b) examined the impact of internal policies and external trade barriers on China's solar PV exports from 2007 to 2016 and found significant spatial pattern differences in PV exports before and after 2011, with exports becoming increasingly concentrated in the Yangtze River Delta region.

ACPs demonstrate both pronounced trade-destruction effects and significant trade-diversion effects. Kuang and Xiang (2019) found that following the imposition of ACPs by the U.S. on solar PV products from mainland China and Taiwan region of China in 2015, mainland China exported to the U.S. declined by 68.1 %, and Taiwan region of China exported fell by 30.6 %. Wang and Xie (2025) found that ACPs imposed by the EU and the U.S. led to a sharp short-term decline of approximately 87.5 % in China's solar PV exports, along with a reduction of about 10.7 % in the number of new market entrants. The economic impact of the EU's "double remedies" on China has been found to be greater than that of the U.S. measures (Wang et al., 2018). These measures not only increased export costs but also weakened firms' capacity for capital accumulation. Zou et al. (2017) investigated the effects of U.S. and EU import restrictions and found that these measures led to severe overcapacity in China's solar PV market, forcing many firms to continuously reduce product prices and driving some into bankruptcy due to intense market competition. Similarly, following the imposition of ACPs, Taiwanese solar PV manufacturers experienced substantial profit declines, with some firms even facing net capital outflows, leading to contractions in production capacity and a shrinking market share (Liu et al., 2016). Conversely, when trade protection measures are eased, investors are more inclined to expand capacity in solar panel production. Pan et al. (2019) indicated that tariff reductions significantly increase returns on investment, lower perceived risks, and encourage firms to adopt more export-oriented strategies, thereby enhancing the overall efficiency of the global industry chain.

Recent studies have documented the emergence of green barriers as non-tariff trade barriers in international trade (De Melo and Solleder, 2020). Several developed countries have implemented compliance requirements addressing the life cycle of solar PV products, including carbon footprint standards (Polverini et al., 2023) and environmental regulations (Weckend et al., 2016). Research indicates that these measures create new challenges for developing country exporters, including China (A. Ali et al., 2024). However, Chen (2024) examined the impact of green trade barriers on China's PV exports to ASEAN from 2010 to 2021 and found that green barriers did not universally hinder exports. Under specific conditions, they may positively influence China's solar PV exports to ASEAN countries. Chen (2024) attributed it to higher environmental thresholds elevating technological standards and enhancing the international reputation of Chinese products, thereby improving their global competitiveness.

However, green and technical barriers have intensified scrutiny from the U.S. and EU regarding China's green subsidy practices. Voituriez and Wang (2015) pointed out that although the China-EU solar PV trade dispute was temporarily resolved through a price undertaking mechanism, the continued provision of green industrial subsidies may trigger renewed conflicts in the future. From a macro-political perspective, the solar PV industry is increasingly viewed by some countries not merely as a category of traded goods, but as a strategic asset. The supply of low-cost energy is directly linked to national security (Apergi et al., 2023). Therefore, solar PV trade barriers also reflect the evolving dynamics of green geo-economic competition among major powers.

4.2. Internal risks classification and interrelationship analysis

According to industry life cycle theory, the evolution of an industry typically follows a standard pattern consisting of four stages: emergence, growth, maturity, and decline (Klepper, 1997). Each stage is characterized by distinct features in terms of competition dynamics (Lenox et al., 2007), technological changes (McGahan and Silverman, 2001), and market behavior (Karniouchina et al., 2013). Some scholars have

examined the developmental stages of China's photovoltaic industry, including the evolution of policies and patent innovations (Sun et al., 2014; Zou et al., 2017), which provide empirical evidence that exemplifies this evolutionary trajectory. Similarly, based on the risk identification analysis presented in Section 3, the study finds that risk profiles vary significantly across different stages of the industry life cycle. Therefore, a temporal perspective is essential for understanding how risks evolve over time and for identifying the dominant risk factors and their interactions in each stage.

As a typical form of external risk, trade frictions are indeed a significant factor affecting China's solar PV industry chain, but they are by no means the sole source of risk (Zhang et al., 2021a, 2021b; Yang et al., 2024; Sun et al., 2024). With increasing globalization and the accelerating pace of technological advancement, internal risks within the solar PV industry chain are exhibiting systemic characteristics. The International Energy Agency (IEA) conducted a comprehensive analysis of the solar PV supply chain from raw materials to finished products, covering five major manufacturing segments: polysilicon, ingots, wafers, cells, and modules. Their analysis examined supply and demand, production, energy consumption, emissions, employment, production costs, investment, trade, and financial performance, highlighting key vulnerabilities and risks at each stage (IEA, 2022). Based on risks identification across different segments of the industry chain in section 3.2, the study categorizes internal risks into six major types: resource risk, technological risk, financial risk, market risk, policy risk, and environmental risk and further analyzes their interrelationships and systemic features.

Scholars have analyzed the implications of resource risks within the solar PV industry chain. Research showed that resource risk in the solar PV industry chain is primarily reflected in the acquisition of critical raw materials and strategic resources (Carrara et al., 2020; Mertens et al., 2024; Y. Zhao et al., 2023), as well as in the volatility of raw material prices (Dustin and Morgan, 2023; Bernreuter, 2025). Resource risks have been shown to influence both technological and financial risks of the industry. Specifically, the impact of critical raw material acquisition risks on technological risks is reflected in how resource constraints limit industrial scale expansion and technology pathway selection. In the early development stage of China's solar PV industry, China's reliance on polysilicon raw materials was relatively high, especially before the domestic production capacity reached scale, with imports mainly from countries such as the U.S, Germany, and Japan (Jiang, 2013). Marigo (2007) found that during the early development stage of China's solar PV industry chain, the key resource risk was silicon shortage and silicon material supply bottlenecks, which directly forced companies to seek silicon resources globally, thereby increasing supply chain risks and technological risks and further constraining the large-scale development of the solar PV industry. On the other hand, the production of silicon materials involved high technological barriers, as well as energy-intensive and costly processes, creating entry barriers that imposed technical and cost pressures on early industry participants (Ye and Zhang, 2014). With technological advancements and industrial expansion, demand for rare metals such as silver, indium, gallium, and tellurium has grown, gradually exposing China's reliance on imports for critical minerals (Behar et al., 2021; Gervais et al., 2023; Zhang et al., 2021a, 2021b; Zhao et al., 2023a, 2023b). According to the IEA's *Net Zero by 2050 Roadmap*, by 2030, silver demand from solar PV manufacturing may exceed 30 % of the total global silver supply in 2020 from the current level of approximately 10 % (IEA, 2021). The shortage of key mineral resources could constrain the choice of technological pathways and heighten technological risk (Lakshmanan et al., 2022; Sverdrup et al., 2024). Furthermore, frequent fluctuations in upstream silicon prices have increased the cost and uncertainty of resource allocation, which in turn affects financial risks. Between 2020 and 2022, the global push toward carbon neutrality, combined with pandemic-related constraints on overseas production capacity, widened the supply-demand gap and triggered a sharp surge in silicon prices (Bernreuter, 2025). However, unstable resource supply has disrupted corporate production planning. Polysilicon prices began to decline continuously from 2023 onward, with some upstream solar PV firms facing costs below cash break-even levels (Casey, 2024; OPIS, 2024). As a risk rooted in the upstream segment of the solar PV industry chain, resource risk affects firms in several ways: falling raw material prices compress profit margins and increase financial risk (Dehghanimadvar et al., 2024).

The evolution of technology risk in the solar PV industry demonstrates distinct phases, transitioning from early technological barriers and equipment dependence to current challenges of rapid iteration and technology pathway selection. In the early stages of China's solar PV industry, polysilicon production technology was primarily monopolized by European and American companies. Core equipment such as crystal pullers and precision coating machines relied heavily on imports, with low domestic production rates, resulting in low production efficiency and high costs (Ye and Zhang, 2014). China's patented inventions in this field were characterized by relatively low average value, and the proportion of R&D investment to revenue remained lower than that of Western counterparts (de la Tour et al., 2011). Since 2015, technical risks have shifted from basic material production to high-end manufacturing technologies and accelerated technological iterations. Although China has made significant progress in manufacturing capacity, it still relies on imports for high-end manufacturing equipment such as silicon wafer cutting machines and polysilicon purification facilities, and has not yet fully localized production (Cai et al., 2019). With the intensification of the U.S. restrictions on China in technology and high-end manufacturing sectors, the technological security has emerged as a key new risk factor (Xu et al., 2022; Sekar et al., 2024). As the pace of technological change intensifies, the solar PV industry must balance investment in technology deployment with research and development (Santen and Anadon, 2016). Between 2023 and 2024, the industry has witnessed a rapid transition from P-type to N-type technologies, with fierce competition among TOPCon, HJT, and BC technologies (Fischer et al., 2021; Ru et al., 2024). According to the China Photovoltaic Industry Development Roadmap (2024–2025) report, newly commissioned mass production lines were primarily N-type cell production lines in 2024. With the rapid release of N-type cell capacity, PERC cell market share declined to 20.5 %, while N-type TOPCon cell market share reached 71.1 %, becoming the dominant cell technology pathway (CPIA, 2025). The speed and scale of technology transition has created greater technology selection pressure and investment decision risks for PV companies, while R&D investment effects demonstrate significant uncertainty and time lag. Existing research showed that R&D expenditure creates a double-edged effect for startups: insufficient spending may result in lost competitive advantages, while excessive spending can become a financial burden, depleting resources and accelerating failure (Feng et al., 2022). The impact of R&D investment on corporate financial performance is not always immediate but involves time lag effects (Teti et al., 2023). Small and medium enterprises (SMEs) face multiple challenges in rapidly evolving technological environments, particularly in technology absorption and business model innovation, due to limited resources and the complexity of technology transition (De Mattos et al., 2024). Consequently, faster technology cycles increase the frequency of market reshuffling, with technology and financial risks intertwining to amplify market concentration risks.

Financial risk in the solar PV industry encompasses financing, cost control, and profit stability. In the early stages, when subsidy policies were still underdeveloped, solar PV projects required high initial investments, had limited financing channels, and faced long payback periods, creating significant cost barriers to market entry (Shuai et al., 2019; Wang, 2014), which is because the application and promotion of PV cells and modules have been partially dependent on financing channels, making market development vulnerable to financial constraints (Sun et al., 2014). Zhang et al. (2012) found that high initial costs and heavy reliance on financing remained major barriers to widespread adoption, particularly affecting the investment decisions of

small-sized and medium-sized users and enterprises, thereby limiting market penetration. In the later stages, financial risk has shifted from entry barriers to profitability challenges. On one hand, the continued decline in the prices of solar cells and modules has severely compressed profit margins, leading even major firms to incur losses and contributing to the systemic spread of financial risk (Yang et al., 2024). On the other hand, external trade barriers, combined with the gradual phasing-out of domestic subsidies, have significantly increased profitability pressures on solar PV enterprises (Cai et al., 2024; Dehghanimadvar et al., 2024).

Market risk in the solar PV industry is primarily reflected in supply–demand imbalances, high market concentration, and excessive dependence on international exports. In the early stages, domestic demand for solar PV products lagged behind global levels, leading to a strong reliance on export markets (Wang et al., 2014). The global financial crisis and subsequent reduction in policy support for solar PV in European countries resulted in a decline in installation demand, severely impacting the export of Chinese solar PV products (Yu et al., 2016). However, domestic production capacity continued to expand rapidly. By 2019, Chinese firms accounted for approximately 72 % of global module manufacturing capacity, and the top ten cell manufacturers controlled nearly 59 % of the market share (Ladislaw et al., 2021). The decline in demand from major export markets, combined with export restrictions and insufficient domestic consumption, led to an oversupply of solar PV materials and equipment. This supply–demand mismatch triggered a vicious cycle of aggressive price competition within the industry (Yu et al., 2021; Guo et al., 2025). By the end of 2021, China's PV module production capacity had reached 359.1 GW, intensifying price competition and even eroding profit margins across the industry (Yang et al., 2024). Furthermore, market concentration has intensified, with leading enterprises increasing their market share, thereby squeezing the survival space for small and medium-sized enterprises (Yang et al., 2024). Data showed the Chinese firms have come to dominate solar PV in the past decade, supplying 80 % of the top ten manufacturers in 2018 (Goldthau and Hughes, 2020). As a result, market risk is bidirectionally coupled with technological, policy, and financial risks, amplifying the overall vulnerability of the solar PV industry.

Policy risks primarily manifest as the early reliance on government subsidies and the later reduction or withdrawal of these subsidies, with policy changes directly impacting the economic risks of the PV industry. Specifically, Dong et al. (2021) found that an increase of 0.1 yuan/kWh in PV subsidies adds about 18 GW/year of installed capacity to the national PV market. However, these subsidies also led to issues such as excessive corporate dependence on government support, blind investment, and low production efficiency (Xiong and Yang, 2016; Yu et al., 2021). On May 31st, 2018, the introduction of the '531 Policy' marked a pivotal turning point. When subsidies are reduced or cancelled, the profitability of PV projects decreases, causing some investors to adopt a wait-and-see attitude, which leads to a reduction in newly installed capacity. Following the implementation of the 531 Policy, China's new PV installed capacity in 2019 was 26.81 GW, a year-on-year decrease of 42.6 % (Lin and Zhang, 2025). Therefore, PV enterprises need to seek new profit growth points, and when combined with financial risks, this creates compounded pressure for solar PV enterprises (Liu et al., 2021; Luan and Lin, 2022). Yan et al. (2019) investigated the impact of government subsidy reductions on the performance of upstream, midstream, and downstream enterprises in the PV industry chain, finding that since 2018, downstream PV enterprises, as the segment most reliant on government subsidies, have been particularly severely affected, experiencing declines in both sales and profitability.

Environmental risks are primarily represented by pollution from early polysilicon production and the environmental impact of later retired solar PV modules. The early environmental risks were evident as the expansion of China's polysilicon industry generated environmental externalities characterized by high energy consumption and severe pollution, particularly the discharge of fluorine-containing wastewater and slurry from cutting processes (Liu et al., 2015). As a large number of PV systems approach their end-of-life, the issue of module recycling is becoming increasingly prominent. Improper disposal of waste PV modules can lead to environmental pollution, such as heavy metal leakage (Pu et al., 2021).

In summary, the systematic literature review has revealed the interconnectedness of internal risk factors within China's PV supply chain and confirmed that these risks do not exist in silos but rather form a highly coupled and intricate system. However, existing literature tends to focus on the impact of one factor on another in isolation. Few studies have treated risks as a holistic entity or examined the cumulative or amplifying effects of the coupling between multiple risk factors on the overall system. Addressing this gap is crucial for a more comprehensive understanding of risk dynamics in the industry.

4.3. *Coupling analysis of systemic risks in China's solar PV industry chain*

Systemic risk coupling refers to the interconnection and mutual influence among different types of risks within the solar PV industry chain, wherein the occurrence or intensification of one type of risk can trigger or exacerbate others. Such coupling effects may amplify the initial risk and result in more severe systemic consequences. Existing studies have emphasized that risks often do not occur in isolation, but instead exhibit interdependent and interactive characteristics (Yim et al., 2015; Zhang, 2016a, 2016b). When a risk arises within an interdependent system, it can affect the system as a whole (De Amorim et al., 2018; Fraume et al., 2020; Qazi and Al-Mhdawi, 2023). This phenomenon has been confirmed in various sectors such as the renewable energy industry (Xu et al., 2015), the oil supply chain (Fazli et al., 2015), and the natural gas sector. The rapid development of China's solar PV industry further illustrates this pattern.

As Qazi et al. (2015) have noted, risks in complex systems are often amplified through cross-interactions and feedback loops, transforming initial shocks into compounded systemic risks. To further investigate how external trade frictions and internal risks across different segments of the solar PV industry chain interact and couple, the study draws on the concepts of risk interdependence (van Dorp and Duffey, 1999) and risk coupling analysis (Yao et al., 2024). Based on these frameworks, a conceptual model illustrating the systemic risk coupling relationships within China's solar PV industry chain was developed, as shown in Fig. 2.

Fig. 2 first identifies a range of external trade frictions, including import tariffs, other forms of import restrictions, government aid and subsidy measures, trade defense instruments, export controls, public procurement and localization policies, and technical barriers to trade, which form strong couplings with six major categories of internal risks within China's solar PV industry chain: resource, technological, market, financial, policy, and environmental risks. This coupling is not a simple linear transmission process but rather the result of the accumulation and interaction of multiple risk factors within the industrial system (Cao et al., 2024). The single, dual, and multiple risk factor couplings depicted in the figure not only reflect the number and combination of risk sources but also indicate varying levels of coupling intensity (Hai et al., 2022). Single-factor coupling typically refers to a system disturbance dominated by one type of risk. Dual-factor coupling involves the resonance and mutual reinforcement between two risk types. Multiple-factor coupling, in contrast, denotes the simultaneous interaction of three or more types of risks, often accompanied by stronger nonlinear feedback and amplification effects, ultimately forming a complex structure of systemic risk (Burkholz et al., 2016; Yao et al., 2024).

Coupled risks propagate through the solar PV industry chain, from upstream raw materials and basic manufacturing to midstream cell and module production and downstream system integration and application, intensifying internal systemic instability. As risks accumulate across spatial and temporal dimensions, the system gradually approaches a

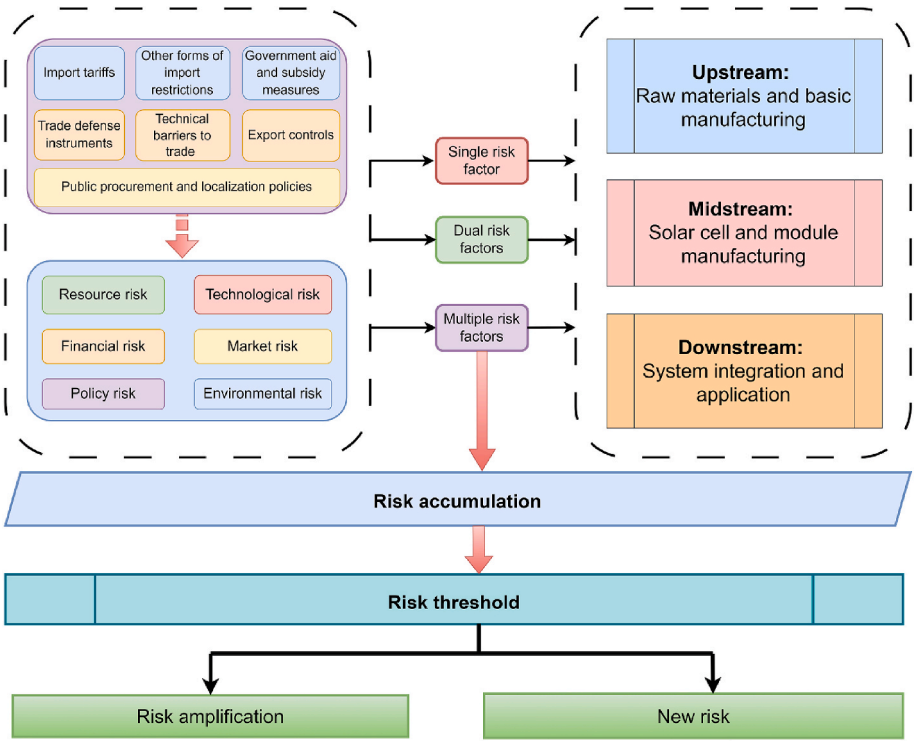


Fig. 2. Systemic risk coupling in China's solar PV industry chain under trade frictions.

critical threshold. Once this threshold is breached, risk coupling can nonlinearly amplify the initial shock, evolving into systemic turbulence and potentially triggering entirely new forms of risk (He et al., 2023). The concept of risk coupling effects has been widely applied in various fields, including engineering projects (Chen et al., 2022), supply chain networks (Wang et al., 2021a, 2021b), and diversified networked industrial chains (Chen et al., 2024). However, its application within the PV industry chain remains relatively scarce.

5. Risk assessment of China's solar PV industry chain

Based on the risk identification and analysis presented in previous sections, the section systematically assesses both external trade friction risks and internal risks confronting China's PV industry chain. Therefore, this section will address two core questions: (1) What are the mainstream methods for risk assessment within the current solar PV industry chain? and (2) How do various types of risks within the industry chain transmit and propagate, leading to systemic risks?

5.1. Risk assessment methods of solar PV industry chain

The core of risk assessment lies in two key aspects: the probability of a risk event occurring, and the consequences and impacts should such an event materialize (Kaplan and Garrick, 1981). As outlined in the ISO 31000:2018 Risk Management Guidelines, risk assessment methods are generally categorized into three types: qualitative, semi-quantitative, and quantitative approaches. Qualitative methods primarily rely on expert judgment and subjective analysis, often utilizing tools such as literature reviews and structured interviews to identify and classify risk factors (Hawkes and Rowe, 2008). Although qualitative methods do not produce numerical outputs, they are widely used for preliminary risk assessment, especially in contexts where empirical data is limited or where expert judgment is essential. Semi-quantitative methods serve as a bridge between qualitative and quantitative approaches by converting subjective judgments into standardized, quantifiable forms, thereby enhancing the systematization and comparability of the assessment (Khorsandi and Aven, 2017). Quantitative methods are based on accessible objective data and employ techniques such as mathematical modeling, probability analysis, and simulation to provide more precise estimates of both the likelihood and consequences of risk events (Sage, 2015). The study systematically reviews and summarizes the main assessment methods currently applied in research on risk assessment within the solar PV industry chain, including the types of risks considered, the specific segments of the industry chain involved, and the particular focus areas of these assessments. Detailed information is presented in Table 2.

Despite considerable progress in the identification and analysis of risks within the solar PV industry chain, existing assessment approaches still exhibit three major limitations. First, most evaluations adopt a static perspective, lacking insight into dynamic transmission mechanisms. Current studies predominantly rely on literature reviews, interviews, or statistical modeling to identify isolated risk factors or make qualitative judgments. Such methods fall short in revealing how risks propagate and evolve across different stages of the industry chain. Given the highly coupled nature of the solar PV industry chain, risks tend to transmit across phases and levels, making static approaches insufficient to capture their dynamic evolution. Second, there is a significant gap in analyzing systemic risks and multi-risk couplings. While the solar PV industry chain is exposed to both external environmental factors and internal structural risks, most existing research focuses on single risk types, overlooking the interactive and co-evolutionary dynamics of multiple risks. Third, current assessments often emphasize outcomes over processes, with limited attention to modeling the mechanisms of risk formation. Many studies focus on quantifying the probability or consequences of risk events while neglecting the antecedent pathways and key nodes involved in risk accumulation.

5.2. Risk transmission in solar PV industry chain

Risk transmission within an industry chain refers to the process by which a risk event occurring at a specific segment or node propagates to other parts of the chain through upstream-downstream relationships,

Table 2
Comparison of risk assessment methods in the solar PV industry chain.

Method Type	Research Approach	Risk Type	Industry Chain Segment	Assessment Focus	Reference
Qualitative	Literature review	Environmental risk, technological risk, regulatory risk, economic risk, social risk	Downstream (solar PV systems)	Consequences and impacts of risk	Dellosa et al. (2024)
	Literature review	Supply–demand imbalance, price risk, technological risk	Upstream (crystalline silicon)	Consequences and impacts of risk	Bye and Ceccaroli (2014)
	Literature review	Supply–demand imbalance, market risk, technological risk, price risk	Midstream (solar PV modules and cells)	Consequences and impacts of risk	Zhao et al. (2015)
	Literature review	Technological innovation risk, market risk	Entire industry chain	Consequences and impacts of risk	Zou et al. (2017)
	Structured interviews	Financial risk, logistics and supply chain risk, policy risk	Downstream (solar PV power projects)	Consequences and impacts of risk	Kayser (2016)
Semi-quantitative	Failure Mode and Effects Analysis (FMEA)	Technological risk	Midstream (solar PV modules)	Probability and consequences of risk	Patil et al. (2024)
	Literature review, exploratory interviews, statistical analysis	Operation and maintenance (O&M) risk	Downstream (solar PV systems)	Probability of risk occurrence	Kut et al. (2024)
	Literature review, fate and transport modeling	Environmental risk (ecological and human health risks)	Downstream (decommissioned solar PV)	Consequences and impacts of risk	Nain and Kumar (2020)
	Literature review, multi-attribute decision-making	Environmental risk, socio-economic risk	Downstream (installation and operation of solar PV stations)	Consequences and impacts of risk	Pimentel Da Silva et al. (2020)
	Literature review, multi-criteria decision-making	Resource security risk	Upstream (critical materials)	Probability and consequences of risk	Y. Zhao et al. (2023)
	System dynamics modeling	Trade frictions	Upstream (polysilicon)	Consequences and impacts of risk	Sandor et al. (2018)
	Failure Mode and Effects Analysis (FMEA)	Investment risk	Downstream (solar PV systems)	Probability and consequences of risk	Rasouli et al. (2023)
Quantitative	Differential sensitivity analysis	Economic risk	Downstream (solar PV power projects)	Consequences and impacts of risk	Guindon and Wright (2020)
	Statistical distribution fitting and uncertainty analysis	Climate risk	Downstream (solar PV power projects)	Probability of risk occurrence	Fernández Peruchena et al. (2016)
	Intensity–area–duration analysis, high-resolution resource data modeling	Climate risk	Downstream (solar PV power projects)	Probability of risk occurrence	Sun et al. (2024)
	Game theory and panel data empirical analysis	Financial risk, policy risk	Entire industry chain (solar PV enterprises)	Consequences and impacts of risk	Xu et al. (2020)
	Project economic modeling	Curtailment risk	Downstream (solar PV power projects)	Consequences and impacts of risk	(O'Shaughnessy et al., 2021)

trade linkages, or technological dependencies (Ouyang et al., 2022). As a fully integrated and highly interconnected system, China's solar PV industry chain spans multiple stages from upstream silicon material supply to midstream module manufacturing and downstream power generation systems, where nodes exhibit strong interdependencies. Consequently, localized disruptions may escalate into systemic shocks (Zhang et al., 2024). The theory of systemic risk provides a conceptual foundation for analyzing the spatial dimension of risk transmission in the solar PV industry chain. This theoretical framework emphasizes how risks propagate through complex interconnections, offering explanations for how localized disturbances can evolve into broader systemic disruptions within the chain (Scheibe and Blackhurst, 2018).

Existing studies on risk transmission within the solar PV industry chain have primarily approached the issue from perspectives such as global trade networks, resource supply, and technological innovation. Ding and Ren (2025) identified the evolutionary characteristics of various segments within the global solar PV trade network and analyzed the transmission of systemic instability by simulating failures at key nodes, highlighting the typical features of risk diffusion. Both the technological structure and geographic dependencies have been recognized as important mechanisms facilitating such diffusion. miao et al. (2024) and Gervais et al. (2023) focused on the upstream resource segment of the solar PV industry chain. miao et al. (2024), using high-purity polysilicon as the core, constructed a multilayer network supply risk propagation model and revealed that the scope of risk propagation expands over time and exhibits a long-tail distribution. Furthermore, a distinct structural asymmetry was observed in the direction of risk transmission between the U.S. and China. The U.S. exerts a strong risk transmission influence on China, while China's impact on the U.S. remains relatively limited. Gervais et al. (2023) examined the global supply chain of silver, a critical material in solar PV production, and found that over 87 % of PV-grade silver must transit through at least one country with a high ESG risk score. This concentration of supply pathways suggests that any disruption at high-risk nodes could trigger downstream transmission effects, from raw material supply to module manufacturing and ultimately to power generation markets, thereby constituting a significant single-point failure risk. This underscores the amplifying role of geopolitical dependency on raw materials in shaping systemic risk. On the technological innovation front, Guo and Li (2024), through a patent network analysis, found that China's innovation resources in the solar PV sector are primarily concentrated in the midstream and downstream segments. The internal connections within the network have become more direct and diversified, although the overall network density is declining.

Overall, although these studies provide a solid foundation for understanding risk diffusion at the global level, they have yet to adequately address how risks are transmitted and evolve within China's domestic industry chain, specifically across upstream raw material supply, midstream manufacturing, and downstream application segments.

5.3. Dynamic risk assessment model

Risk transmission is not a static event but a temporal evolution process, unfolding through sequential stages that illustrate how risk emerges, spreads, and ultimately causes impact. This process begins with the accumulation of preconditions, followed by the triggering of a risk event, which subsequently forms a risk footprint and leads to impact through a propagation mechanism (Buldyrev et al., 2010; Helbing, 2013). Keow Cheng and Hon Kam (2008) proposed a conceptual framework for analyzing risk in supply networks. Within this framework, "Preconditions" refer to the vulnerabilities of certain nodes within the network, defined as characteristics that make specific parts of the system more susceptible to shocks. Once these preconditions exceed a critical threshold, a "Risk Event" is triggered. The term "Risk Footprint" describes the collection of risk-affected nodes in the network resulting from such events. Directional arrows clearly indicate the path of risk propagation or transfer throughout the system. Referring to the framework and risk analysis in Section 4.2, the dynamic risk transmission model of China's solar PV industry chain is illustrated in Fig. 3.

Fig. 3 shows the dynamic risk transmission model within China's PV industry chain, characterizing the risk transmission mechanism from both temporal and spatial dimensions. From the temporal perspective, risks exhibit horizontal evolution characteristics, following a sequential development path of "Preconditions → Risk Event → Risk Footprint → Propagation → Impact." When vulnerability characteristics (preconditions) of certain nodes in the solar PV accumulate and exceed critical thresholds, risk events are triggered. Subsequently, these risk events form risk footprints within specific areas, which represent the collection of risk-affected nodes in the PV industry, and ultimately generate systemic impacts through propagation mechanisms. The temporal evolution process reflects the dynamic transformation of risks from latent states to explicit hazards within the solar PV industry chain. From the spatial dimension, the model identifies multiple transmission modes of risks across different tiers of the PV industry chain. Specifically, risk propagation encompasses four primary pathways: same-layer forward propagation (red solid arrows), cross-layer forward propagation (red dashed arrows), cross-layer backward propagation (blue dashed arrows), and circular propagation (purple dashed arrows). Additionally, external trade friction serves as a significant risk source that can directly impact all tiers of the solar PV supply chain (green arrows).

6. Risk response strategies

In the field of risk management, four primary response strategies are widely recognized: risk avoidance, risk acceptance, risk transfer and cooperation, and risk insurance (Fazli et al., 2015). Risk avoidance refers to the termination of, or decision not to engage in, activities that may lead to unacceptable levels of risk (Eckerd, 2014). Risk acceptance involves the deliberate decision to retain a certain level of risk, typically based on cost-benefit considerations or strategic constraints (Aven and Vinnem, 2005). Risk transfer and cooperation entail shifting potential losses to a third party through mechanisms such as insurance or outsourcing (Cronk et al., 2019). Risk mitigation, the most widely used approach, aims to lower the likelihood or potential impact of adverse events through preventive or adaptive strategies (Talluri et al., 2013). These response strategies provide a useful framework for analyzing how China's PV industry has addressed internal and external risks across its four key developmental stages.

In the initial stage, China's PV industry was in the nascent phase of technological development. Due to the absence of strong domestic demand, early risk treatment efforts primarily relied on risk acceptance, whereby stakeholders consciously tolerated certain levels of market and technological uncertainty due to strategic constraints and limited alternatives. de la Tour et al. (2011) conducted interview-based research to investigate the development drivers of China's solar PV industry and found that the industry initially acquired production capabilities through multiple channels: the import of advanced manufacturing equipment, the return of skilled entrepreneurs from the Chinese diaspora who brought critical know-how, and technology spillovers from foreign-invested firms. These approaches allowed China to rapidly enter the manufacturing segment while minimizing early-stage R&D expenditures. On the other hand, the sector adopted a risk mitigation strategy by strategically targeting export markets, which helped mitigate the limitations of domestic market immaturity and generated essential revenue streams (Dunford et al., 2013). These measures reflect China's early efforts to treat risk through technology acquisition and integration into international markets.

The rapid growth stage marked by the success of China's PV exports, was accompanied by international trade frictions, including anti-dumping cases, countervailing duties, and import restrictions from major markets. In response, the industry adopted a mixed strategy of risk mitigation and partial risk transfer. In the face of anti-dumping measures

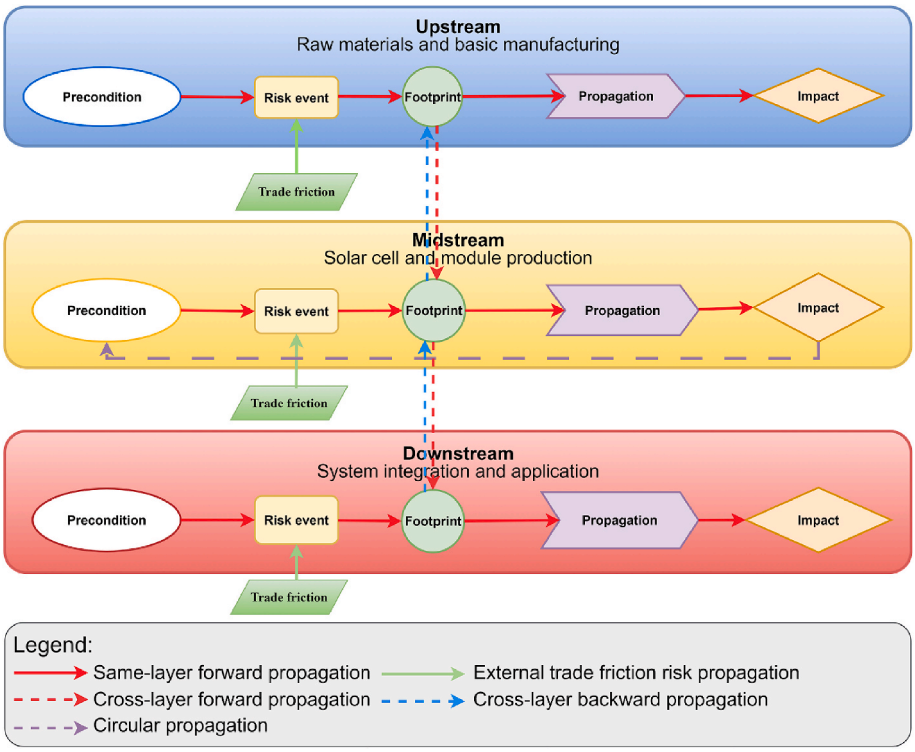


Fig. 3. Dynamic risk transmission model of China's solar PV industry chain.
Source: developed by authors basing on the study (Keow Cheng and Hon Kam, 2008).

imposed by the U.S. and the EU, the Chinese government engaged in diplomatic negotiations through WTO (Hajdukiewicz and Pera, 2020) while simultaneously introducing domestic subsidy programs to stimulate internal demand and reduce reliance on exports, such as the "Golden Sun Program" (Zhi et al., 2014). Although diplomatic efforts achieved limited success, the strategies toward domestic demand provided a crucial buffer against the decline in exports. In addition, vertical integration emerged as a key strategy, as firms consolidated upstream, midstream, and downstream operations to stabilize supply chains and mitigate price volatility (Zhang and Gallagher, 2016).

During the maturity and adjustment period, China's PV industry faced significant challenges, including severe overcapacity, subsidy reforms, and ongoing trade restrictions. As both government and enterprises worked to restructure the industry, risk mitigation emerged as the dominant strategy. Firstly, domestic market expansion was prioritized through feed-in tariff programs (Ye et al., 2017) and rural solar initiatives such as photovoltaic poverty alleviation projects (Bai et al., 2024a), which helped mitigate the impact of declining exports. Secondly, by redirecting exports to emerging markets such as Vietnam, Malaysia through overseas manufacturing hubs and localized project partnerships (Zhu et al., 2021b), Chinese PV companies have leveraged Belt and Road Initiative (BRI) markets as a buffer zone to mitigate the impact of international trade sanctions (Zhu et al., 2024). As a result, China's solar industry reduced dependence on Western markets, mitigating U.S./EU anti-dumping tariffs (Liu and Shi, 2019; Jackson et al., 2021). Thirdly, the Chinese government issued successive adjustment and transformation policies. In 2013, the State Council issued Some Suggestions on Promoting the Healthy Development of Photovoltaic Industry by setting production standards and promoting industrial consolidation to curb overcapacity (Hu et al., 2020). Finally, enterprises intensified R&D investments to advance PERC cells, monocrystalline silicon technologies, and diamond wire cutting processes, boosting solar cell efficiency while slashing costs through material optimization and localized equipment manufacturing, thereby mitigating technological bottlenecks and cost volatility risks (Lv et al., 2020).

The globalization and transformation period marks a strategic shift of Chinese PV enterprises toward full internationalization and technological leadership, driven by the increasing complexity of trade protectionism. During this phase, the PV industry has integrated risk transfer and risk avoidance strategies on a global scale. Major Chinese PV manufacturers have actively expanded overseas production capacities, particularly in Southeast Asia and the Middle East, to circumvent trade barriers and meet localized demand (Rizqullah Hasian Rambey et al., 2022; Lin and Liu, 2025). Sustained investment in advanced technologies such as TOPCon, heterojunction, and perovskite solar cells has helped Chinese firms maintain global competitive advantages while reducing reliance on price competition (Md.F. Ali et al., 2024; Ghosh et al., 2022). Supply chains have been strategically diversified both geographically and operationally to mitigate risks from geopolitical tensions and raw material constraints (Wang and Cheng, 2025). Furthermore, the adoption of integrated business models has provided new revenue streams, including building-integrated photovoltaics (BIPV) (Peng et al., 2011), agro-photovoltaic systems (Xia et al., 2024), floating PV plants (Bai et al., 2024b), solar-powered transportation infrastructure, and desert remediation projects (Hernandez et al., 2019).

7. Conclusions and research prospects

7.1. Conclusions

This study follows the framework of risk management theory to systematically analyze the dynamic risks and strategies within China's solar PV industry chain under trade frictions.

(1) To answer the first research question, "What types of risks have emerged across different segments of China's solar PV industry chain during its development, particularly under the impact of trade frictions, and in what ways are these risks dynamic?", the risk identification section categorizes trade barrier risks into six types and identifies the main risks in China's PV industry chain as

resource, technology, market, financial, policy, and environmental risks through bibliometric analysis and literature review. The risk analysis section systematically examines the complex interrelationships among these identified risks through a comprehensive literature review. For instance, exiting studies indicate that upstream resource acquisition risk influences technology risk (Ye and Zhang, 2014). Price fluctuations in upstream resources affect the financial risk of midstream PV enterprises (Casey, 2024; OPIS, 2024). During critical early financing periods, enterprise financial risk impacts technology risk; conversely, during periods of rapid technological advancement, technology risk affects enterprise financial risk (Zhang et al., 2012). Furthermore, downstream policy changes differentially impact the financial risks of upstream, midstream, and downstream enterprises (Yan et al., 2019). These findings highlight that the interdependencies among risks in China's PV industry chain are not simple one-way relationships but form a complex risk transmission network. However, existing literature largely focuses on the impact of one factor on another, rarely adopting a systemic perspective on industry chain risks or considering them from a risk network viewpoint. Therefore, the study constructs a conceptual model of systemic risk coupling for China's PV industry chain under trade barrier risks, offering a novel perspective for a comprehensive understanding of industry chain risks.

(2) To address the second research question, "Why is it necessary to assess dynamic risks, and how can they be evaluated?", the risk assessment section emphasizes the inadequacy of traditional static risk assessment methods for the rapidly evolving and dynamic PV industry. Existing dynamic risk assessment approaches, while focusing on a global scale, have a gap in addressing systemic risk transmission within China's PV industry chain, particularly by integrating trade barrier risks. This study innovatively constructs a dynamic risk transmission model for China's solar PV industry chain that incorporates both temporal and spatial dimensions. Temporally, it traces the evolution of risks from latent preconditions to risk events, then to risk footprints, and finally to systemic impacts. Spatially, it identifies multiple transmission pathways across different tiers of the industry chain, revealing interactions that extend beyond a simple linear process. Although the dynamic risk transmission model is presented in a linear form for clarity, it reflects iterative, evolving, and multidimensional dynamics, distinguishing it from traditional static assessments.

(3) To respond to the third research question, "What strategies has China adopted in response to these dynamic risks?", the risk response section identifies that China's PV industry has faced distinct dominant risks at various development stages and has consequently adopted adaptive strategies. In the early development stage, technological import and acquisition were employed to overcome barriers and target export markets. During the rapid growth phase, in response to trade barriers, the government initiated diplomatic negotiations and gradually reoriented subsidy directions towards domestic demand to reduce export reliance, with vertical integration emerging as a core strategy. In the mature adjustment phase, policies were introduced to expand the domestic market, enterprises intensified R&D efforts, export markets diversified, and industry consolidation was promoted to address overcapacity. Entering the globalization transition period, proactive expansion of overseas production capacity to circumvent trade barriers, increased technological R&D, and diversification of application models have become crucial measures. These evolution strategies reflect China's PV industry's learning and adaptive capacity in managing systemic risks.

7.2. Research prospects

Based on the findings of this study and the identified literature gaps, the following future research directions can be pursued:

Deepening the analysis of risk coupling mechanisms: While existing literature analyzes various risks within China's solar PV industry chain, these studies often examine the impact of individual risks on another risk in isolation and rarely investigate the Chinese PV industry chain risks as an interconnected system (Cai et al., 2019; Yu et al., 2021; Yang et al., 2024; Guo et al., 2025). There is a gap in understanding the complex interplay between external trade barrier risks (e.g., import tariffs, TBT) and internal industry chain risks (including resource supply and demand, technological iteration, market fluctuations, and financial pressures), especially in treating these as a holistic, coupled system and adequately analyzing the interdependent effects of various risk combinations on the overall system. Future research should prioritize quantitative assessment under single-factor, two-factor, and multi-factor coupling scenarios. By incorporating advanced quantitative methods such as complex network analysis, system dynamics, and machine learning, studies can precisely measure how different types of trade barriers interact with internal risks. It will allow for the quantification of how these coupling effects amplify or mitigate impacts across various industry chain segments, helping to reveal non-linear superposition effects and potential tipping points of risk, thereby providing a scientific basis for developing more targeted and proactive risk management strategies.

Quantifying and simulating risk transmission paths: Although the study constructs a risk transmission model, the specific transmission paths and intensities among its elements have not yet been quantitatively measured. Most existing studies are based on the perspective of global trade flows, with limited integration of external trade barrier risks and internal risks within China's PV industry chain (Gervais et al., 2023; miao et al., 2024). Therefore, there is a gap in quantitatively identifying the transmission pathways between these external and internal risks. Future research could utilize quantitative methods such as complex network analysis, system dynamics modeling, and agent-based modeling to precisely simulate and predict the diffusion effects of external trade frictions throughout the industry chain. This will provide more actionable insights for risk management decisions by moving beyond conceptual models to empirically testable simulations.

Evaluating and optimizing the effectiveness of strategies: Existing studies often describe strategies qualitatively, but rarely provide empirical evidence of their effectiveness or optimal applicability (Zhang and Gallagher, 2016; Zhi et al., 2014; Zhu et al., 2024). Future research could employ case studies, empirical analyses, or optimization models to conduct comparative analyses of different strategies in response to specific risks. This would also explore how to build more flexible and adaptable risk management systems, providing practical guidance for industry stakeholders.

CRediT authorship contribution statement

Yujia Zhao: Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Benjamin Craig McLellan:** Writing – review & editing, Methodology. **Chaofan Wang:** Writing – review & editing, Visualization. **Jing Shuai:** Funding acquisition. **Wanting Xiang:** Writing – review & editing, Resources, Investigation, Conceptualization. **Chuanmin Shuai:** Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization.

Declaration of generative AI in scientific writing

During the preparation of this work, the authors used ChatGPT to improve the language and readability of the manuscript. After using this tool, the authors carefully reviewed and edited the content as needed

and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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